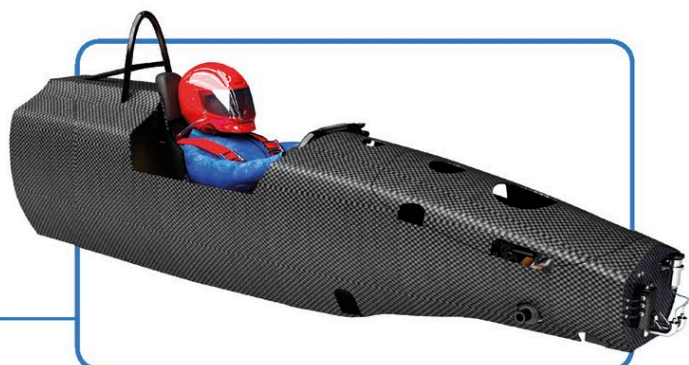




Survival cell

The monocoque, or survival cell, surrounds the cockpit where the driver sits. It is made of a strong, stiff carbon-fiber composite that can absorb the full energy of an impact without being damaged. Carbon fiber is much lighter than steel or aluminium, helping the car go faster and use less fuel.

Mimicking nature

Many synthetic materials were invented to replace natural materials that were too hard, or too expensive, to extract or harvest. For example, nylon was invented to replace silk in fabrics, and polyester fleece can be used instead of wool. Ever advancing technology makes it possible to imitate some amazing materials, such as spider silk, which is tougher than Kevlar®, stronger than steel, yet super flexible.



Steering

A carbon-fiber steering wheel fits into a carbon-titanium column, designed to deform on impact.

Wheel

The wheels are made from one piece of lightweight magnesium alloy under an 11,000-ton press. Alloys are man-made (but not synthetic) materials, produced by mixing metals with metals or other elements (see pp.62-63).

Suspension

Many parts of the suspension system are made of carbon fibers, which are aligned so that the structure is very strong.

Bodywork

An ultra-light layer of strong carbon fiber so thin you can see through it reinforces the car's body.

Tires

Kevlar® and carbon fibers are layered with reinforced rubber compounds. Different compounds are used to cope with different track conditions.

